

Title: Smart Modular Hydroponic System

Abstract:

This invention discloses a smart modular hydroponic system comprising stackable hexagonal growing units with integrated LED grow lights, nutrient reservoirs, and water pumps. Each unit communicates with a central hub employing a machine learning algorithm to optimize growth parameters based on sensor data and user inputs. A mobile application provides real-time monitoring, growth analytics, and maintenance alerts. The system allows for plug-and-play expansion and adaptable nutrient blending.

Field of the Invention:

This invention relates to the field of hydroponic systems, particularly to modular, smart hydroponic systems incorporating sensor technology, machine learning, and mobile application control.

Background of the Invention:

Traditional hydroponic systems often lack scalability, automation, and real-time monitoring capabilities. Existing systems may require significant recalibration with expansion, lack sophisticated control over environmental factors, and offer limited data-driven insights into plant growth. The need for a more efficient, scalable, and data-driven hydroponic system is evident.

Summary of the Invention:

The present invention addresses these limitations by providing a smart modular hydroponic system. This system utilizes stackable hexagonal growing units, each equipped with integrated LED grow lights, a nutrient reservoir, a water pump, and multiple sensors for monitoring various parameters. A central hub collects data from the growing units and employs a machine learning algorithm to optimize growth parameters. A mobile application facilitates user interaction, providing real-time monitoring, growth analytics, and maintenance alerts. The system's modular design allows for plug-and-play expansion without recalibration, while an adaptive nutrient blending module ensures customized nutrient delivery.

Detailed Description:

The smart modular hydroponic system comprises a plurality of stackable hexagonal growing units. Each unit is self-contained, including integrated LED grow lights for optimal light exposure, a nutrient reservoir to hold the nutrient solution, and a water pump for circulating the solution. Multiple sensors within each unit monitor parameters such as pH, nutrient levels, and humidity. The growing units are designed for plug-and-play expansion, allowing users to add or remove units without affecting the system's operation. The units communicate wirelessly with a central hub.

The central hub collects data from the sensors in each growing unit and processes this information using a machine learning algorithm. This algorithm optimizes at least one parameter selected from the group consisting of light cycles, nutrient delivery, and water flow, adapting to real-time conditions and user inputs received via a mobile application. The system also incorporates an adaptive nutrient blending module for mixing customized nutrient solutions for multiple plants within shared reservoirs.

A mobile application provides a user interface for controlling and monitoring the system. The application receives data from the central hub, providing real-time monitoring of various parameters, growth analytics based on the collected data, and maintenance alerts when necessary. The analytics and alerts can be powered by a cloud-based AI model for enhanced processing capabilities.

Claims:

1. A modular hydroponic system comprising: a plurality of stackable hexagonal growing units, each unit including integrated LED grow lights, a nutrient reservoir, and a water pump; and a central hub communicatively coupled to said growing units for receiving data therefrom.
2. The system of claim 1, wherein said growing units are configured for plug-and-play expansion without system recalibration.

3. The system of claim 1, wherein each growing unit further comprises a plurality of sensors for monitoring at least one parameter selected from the group consisting of pH, nutrient levels, and humidity.
4. The system of claim 3, wherein said central hub comprises a machine learning algorithm configured to optimize at least one parameter selected from the group consisting of light cycles, nutrient delivery, and water flow based on real-time sensor data received from said growing units.
5. The system of claim 4, wherein said machine learning algorithm further utilizes user inputs received via a mobile application.
6. The system of claim 1, further comprising an adaptive nutrient blending module for mixing customized nutrient solutions for multiple plants within shared reservoirs.
7. A mobile application for controlling and monitoring a modular hydroponic system, said application configured to: receive data from said system; provide growth analytics based on said data; and provide maintenance alerts based on said data.
8. The mobile application of claim 7, wherein said analytics and alerts are powered by a cloud-based AI model.
9. A method for operating a modular hydroponic system, comprising: monitoring at least one parameter selected from the group consisting of pH, nutrient levels, and humidity within a plurality of growing units using a plurality of sensors; transmitting said sensor data to a central hub; optimizing at least one parameter selected from the group consisting of light cycles, nutrient delivery, and water flow using a machine learning algorithm based on said sensor data and user inputs; and providing growth analytics and maintenance alerts via a mobile application based on said sensor data.
10. The method of claim 9, wherein said user inputs are provided via a mobile application.
11. The system of claim 1, wherein said hexagonal growing units are stackable in a plurality of

configurations.